INCH-POUND
MIL-DTL-24643/58E
22 August 2002
SUPERSEDING
MIL-C-24643/58D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2CS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, size AWG Size No. 26,7 strands.
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Crosslinked polyethylene insulation. Nominal wall thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Standard identification code applied by method 6.
Fourth -	Two conductors (one black and one white) cabled together to form a pair in accordance with table I.
Fifth -	The specified number of pairs (see table II) cabled together in accordance with table III. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.
Sixth -	Binder tape applied helically with overlap.
Seventh -	Braided shield of tin coated copper.
Eighth -	Spirally wrapped 0.002 inch thick polyethylene (polyester film) separator tape with 25 percent minimum overlap.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.

Construction (non-watertight) (continued)

Ninth - Braided shield of tin coated copper.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. (see table III for thickness).

TABLE I. Pair lay.

Left hand lay										
///////					65	66	67	62,68 71,74,77	63,69 72,75	61,64 70,73, 76	PAIR NUMBER
//////	///////				53	54	55	45,48 51,56,59	43,46, 49 52,56, 60	44,47 50,58	
/////	//////	///////			27,39	28,40	29,41	19,22,25 30,33,36, 42	20,23, 26 31,34, 37	21,24 32,35, 38	
////	/////	//////	///////		13 1	14 2	15 3	7,10,16 4	8,11,17 5	9,12,18 6	
LS2CS-77	LS2CS-60	LS2CS-42	LS2CS-18	LS2CS-6	1.6	3.5	2.5	4.5	2.1	2.9	
					LENGTH OF LAY (INCHES)						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.

TABLE II. Details.

Military part no. M24643/58	Type and size	Number of pairs	Cable jacket thickness min avg (inch)	Dia over shielded core (inch nom.)	Overall diameter (inch	
					min	max
-01UD	LS2CS-6	6	0.063	0.300	0.40	0.43
-02UD	LS2CS-18	18	0.063	0.435	0.55	0.59
-03UD	LS2CS-42	42	0.063	0.610	0.75	0.80
-04UD	LS2CS-60	60	0.082	0.790	0.88	0.93
-05UD	LS2CS-77	77	0.085	0.860	1.00	1.07

TABLE III. Cable lay.

	LS2CS-6		LS2CS-18		LS2CS-42		LS2CS-60		LS2CS-77	
	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)
Core	6	3.50	6	3.50	1	1.60	6	3.50	3	2.75
Layer 1	-	--	12	6.00	7	4.00	12	6.50	9	5.00
Layer 2	-	--	-	--	14	6.50	18	8.00	15	7.50
Layer 3	-	--	-	--	20	8.50	24	10.00	22	10.00
Layer 4	-	--	-	--	-	--	-	--	28	12.00

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - ohms/1000 feet at 25°C, maximum 42.17
Voltage withstand - volts, root mean square, minimum
Conductor to conductor 2000
Insulation resistance - Megohms -1000 feet minimum
Conductor to shield 500
Shield to Shield (outer shields only) 100
Jacket flaws ..... No failure
Conductor and shield continuity ..... No failure

Group A:
Visual and dimensional ..... No failure
Characteristic impedance - at 1MHz, ohms ..... 135 ± 10

Group B:
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (when tested at 200°C) 50
Cold bending, cable - mandrel diameter, 12 times cable diameter ..... No damage
Bending endurance - at 25 ± 2°C, cycles, minimum 25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.

Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

EXAMINATION AND TESTS:

Group C:
Physicals (aged)
Insulation
Air Oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum.....125
Shield - conformance to material, construction and coverage..... No failure
Surface transfer impedance
Milliohms per meter, maximum 700
EMP response time - dB, minimum.....60

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)(125 $\pm$ 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Shield insulation 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.

Halogen content - percent, maximum
- Jacket 0.2
- Fillers 0.2

Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet (nominal).

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-056)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.